BuyerRadar report: Solar panels

Query: solar panels
Industry: Solar panels (`solar-panels`)

Photovoltaic cells assembled in modules or made up into panels, sold to electrical wholesalers, solar developers, and EPCs.

How the scores work

Each channel is scored from 1 to 5 on Money (budget), Authority (who can say yes), and Need (fit with this product). The total is the sum, out of 15.

Ties break by Money, then Authority, then Need, then channel type (importing country, buyer type, trade show, job title, marketplace, RFQ board, community), then name.

Scores on curated channels are editorial judgments explained in the reason and tied to a public page (a NAICS title, an O*NET occupation, a fair, a marketplace, or a community). They are not measurements.

Importing-country Money is not editorial. Rank 1 in the Comtrade response scores 5, ranks 2-3 score 4, ranks 4-10 score 3. Authority is 4 because importers in that country hold the order. Need is 5, 4, or 3 according to whether the HS heading is specific, close, or a broad basket. The dollar figure is the unmodified primaryValue literal.

Trade data

HS 854143 is photosensitive semiconductor devices, photovoltaic cells assembled in modules or made up into panels. Loose cells and modules under older codes are not this heading.

- HS code: 854143
- Official description: Electrical apparatus; photosensitive semiconductor devices, photovoltaic cells assembled in modules or made up into panels
- Flow: Import (flowCode M), partner World
- Period: 2024
- Retrieved: 2026-10-27
- Reporters stored: 132
- API count: 132
- Preview complete: yes, count matched stored rows and was under 500
- Query: https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc
- API documentation: https://uncomtrade.org/docs/un-comtrade-api/
- Database: https://comtradeplus.un.org/
- HS nomenclature: World Customs Organization - HS Nomenclature 2022 edition - https://www.wcoomd.org/en/topics/nomenclature/instrument-and-tools/hs-nomenclature-2022-edition.aspx

Ranked channels

The free tier of the site shows the first 5. This file is the full list, with an outreach angle on every channel.

1. Electrical Apparatus and Equipment, Wiring Supplies, and Related Equipment Merchant Wholesalers

- Type: Buyer company type
- Money: 5 / 5
- Authority: 4 / 5

- Need: 5 / 5
- Total: 14 / 15
- Link: https://www.census.gov/naics/?input=423610&year=2022&details=423610

NAICS 423610 wholesales electrical apparatus, which is how modules reach installers. Money is 5 because distributors buy containers. Authority is 4. Need is 5 because a panel is electrical apparatus of the kind this industry stocks.

Outreach: Send watt class, dimensions, the standards you can document, and current stock. Ask which inverter brands their installers already use, and do not promise a certification you cannot show.

Source: U.S. Census Bureau, NAICS 2022, 423610 - Electrical Apparatus and Equipment, Wiring Supplies, and Related Equipment Merchant Wholesalers - https://www.census.gov/naics/?input=423610&year=2022&details=423610

2. Intersolar Europe

- Type: Trade show
- Money: 5 / 5
- Authority: 4 / 5
- Need: 5 / 5
- Total: 14 / 15
- Link: https://www.intersolar.de/home

Intersolar Europe (intersolar.de) is the solar trade fair. Money is 5. Authority is 4 because distributors and developers attend. Need is 5 because modules are a core product of the show.

Outreach: Book distributors for stock business and developers for projects, as two calendars. Bring the datasheet and the warranty terms you will sign.

Source: Intersolar Europe - https://www.intersolar.de/home

3. USA - reported imports of HS 854143

- Type: Importing country
- Money: 5 / 5
- Authority: 4 / 5
- Need: 5 / 5
- Total: 14 / 15
- Link: https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc
- primaryValue USD: 15193756743.0
- primaryValue with separators: 15,193,756,743.0
- Quantity: 103775332.0 u
- Rank among reporters: 1 of 132 in the stored Comtrade response

USA reported primaryValue 15193756743.0 USD (shown with separators as 15,193,756,743.0) for HS 854143 imports from the world in 2024. That number is the UN Comtrade public-preview CIF primary value for partner World. Reported quantity: 103,775,332.0 u (quantity not flagged as estimated). Money is 5 because this reporter ranks 1 of 132 reporters in that response by primaryValue. Authority is 4 because the country is where the buyers are: importers and distributors there hold the purchase order, and the country itself does not. Need is 5. The official heading names this product, so the import value is a direct Comtrade measure of recorded trade in it. Official commodity description: Electrical apparatus; photosensitive semiconductor devices, photovoltaic cells assembled in modules or made up into panels. The

preview count matched the rows stored (132), under the 500-row cap.

Outreach: Do not email a country. Use USA as a map: find importers there that already clear HS 854143 and open with the Comtrade fact that USA reported primaryValue 15193756743.0 USD of this heading in 2024. Offer a first-container spec sheet, not a market-size story.

Source: UN Comtrade public preview API, HS 854143, 2024 imports from World - https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc

4. Purchasing Agents, Except Wholesale, Retail, and Farm Products

- Type: Job title
- Money: 4 / 5
- Authority: 5 / 5
- Need: 5 / 5
- Total: 14 / 15
- Link: https://www.onetonline.org/link/summary/13-1023.00

O*NET 13-1023.00 purchasing agents buy equipment for companies and distributors. Money is 4. Authority is 5. Need is 5.

Outreach: Send the datasheet, the lead time, and the warranty. Ask who signs off on the approved-vendor list.

Source: O*NET OnLine 13-1023.00 - Purchasing Agents, Except Wholesale, Retail, and Farm Products - https://www.onetonline.org/link/summary/13-1023.00

5. Solar Electric Power Generation

- Type: Buyer company type
- Money: 5 / 5
- Authority: 4 / 5
- Need: 4 / 5
- Total: 13 / 15
- Link: https://www.census.gov/naics/?input=221114&year=2022&details=221114

NAICS 221114 is solar electric power generation. Money is 5 because a plant buys modules at project scale. Authority is 4 because procurement and the EPC share the decision. Need is 4 because generators buy modules, often through an EPC rather than from a factory email.

Outreach: Ask who the EPC and the lender are. Factories that cannot meet the lender's equipment list should say so early.

Source: U.S. Census Bureau, NAICS 2022, 221114 - Solar Electric Power Generation - https://www.census.gov/naics/?input=221114&year=2022&details=221114

6. Brazil - reported imports of HS 854143

- Type: Importing country
- Money: 4 / 5
- Authority: 4 / 5
- Need: 5 / 5
- Total: 13 / 15
- Link: https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc
- primaryValue USD: 2875444624.0
- primaryValue with separators: 2,875,444,624.0

- Quantity: 330796215.771 u (estimated flag)
- Rank among reporters: 2 of 132 in the stored Comtrade response

Brazil reported primaryValue 2875444624.0 USD (shown with separators as 2,875,444,624.0) for HS 854143 imports from the world in 2024. That number is the UN Comtrade public-preview CIF primary value for partner World. Reported quantity: 330,796,215.771 u (Comtrade flagged the quantity as estimated). Money is 4 because this reporter ranks 2 of 132 reporters in that response by primaryValue. Authority is 4 because the country is where the buyers are: importers and distributors there hold the purchase order, and the country itself does not. Need is 5. The official heading names this product, so the import value is a direct Comtrade measure of recorded trade in it. Official commodity description: Electrical apparatus; photosensitive semiconductor devices, photovoltaic cells assembled in modules or made up into panels. The preview count matched the rows stored (132), under the 500-row cap.

Outreach: Do not email a country. Use Brazil as a map: find importers there that already clear HS 854143 and open with the Comtrade fact that Brazil reported primaryValue 2875444624.0 USD of this heading in 2024. Offer a first-container spec sheet, not a market-size story.

Source: UN Comtrade public preview API, HS 854143, 2024 imports from World - https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc

7. India - reported imports of HS 854143

- Type: Importing country
- Money: 4 / 5
- Authority: 4 / 5
- Need: 5 / 5
- Total: 13 / 15
- Link: https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc
- primaryValue USD: 2875192304.711
- primaryValue with separators: 2,875,192,304.711
- Rank among reporters: 3 of 132 in the stored Comtrade response

India reported primaryValue 2875192304.711 USD (shown with separators as 2,875,192,304.711) for HS 854143 imports from the world in 2024. That number is the UN Comtrade public-preview CIF primary value for partner World. Comtrade did not return a usable quantity unit for this row. Money is 4 because this reporter ranks 3 of 132 reporters in that response by primaryValue. Authority is 4 because the country is where the buyers are: importers and distributors there hold the purchase order, and the country itself does not. Need is 5. The official heading names this product, so the import value is a direct Comtrade measure of recorded trade in it. Official commodity description: Electrical apparatus; photosensitive semiconductor devices, photovoltaic cells assembled in modules or made up into panels. The preview count matched the rows stored (132), under the 500-row cap.

Outreach: Do not email a country. Use India as a map: find importers there that already clear HS 854143 and open with the Comtrade fact that India reported primaryValue 2875192304.711 USD of this heading in 2024. Offer a first-container spec sheet, not a market-size story.

Source: UN Comtrade public preview API, HS 854143, 2024 imports from World - https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc

8. Electrical Engineers

- Type: Job title

- Money: 3 / 5
- Authority: 4 / 5
- Need: 5 / 5
- Total: 12 / 15
- Link: https://www.onetonline.org/link/summary/17-2071.00

O*NET 17-2071.00 is Electrical Engineers. Money is 3 because they specify more often than they pay. Authority is 4 because a module that fails their design will not be bought. Need is 5 because a panel is an electrical device they size.

Outreach: Send electrical characteristics and dimensions. Ask for the string design constraints. Leave price for purchasing.

Source: O*NET OnLine 17-2071.00 - Electrical Engineers - https://www.onetonline.org/link/summary/17-2071.00

9. Germany - reported imports of HS 854143

- Type: Importing country
- Money: 3 / 5
- Authority: 4 / 5
- Need: 5 / 5
- Total: 12 / 15
- Link: https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc
- primaryValue USD: 2061623089.467
- primaryValue with separators: 2,061,623,089.467
- Rank among reporters: 5 of 132 in the stored Comtrade response

Germany reported primaryValue 2061623089.467 USD (shown with separators as 2,061,623,089.467) for HS 854143 imports from the world in 2024. That number is the UN Comtrade public-preview CIF primary value for partner World. Comtrade did not return a usable quantity unit for this row. Money is 3 because this reporter ranks 5 of 132 reporters in that response by primaryValue. Authority is 4 because the country is where the buyers are: importers and distributors there hold the purchase order, and the country itself does not. Need is 5. The official heading names this product, so the import value is a direct Comtrade measure of recorded trade in it. Official commodity description: Electrical apparatus; photosensitive semiconductor devices, photovoltaic cells assembled in modules or made up into panels. The preview count matched the rows stored (132), under the 500-row cap.

Outreach: Do not email a country. Use Germany as a map: find importers there that already clear HS 854143 and open with the Comtrade fact that Germany reported primaryValue 2061623089.467 USD of this heading in 2024. Offer a first-container spec sheet, not a market-size story.

Source: UN Comtrade public preview API, HS 854143, 2024 imports from World - https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc

10. Netherlands - reported imports of HS 854143

- Type: Importing country
- Money: 3 / 5
- Authority: 4 / 5
- Need: 5 / 5
- Total: 12 / 15
- Link: https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc

- primaryValue USD: 1930661109.952
- primaryValue with separators: 1,930,661,109.952
- Quantity: 22718291.838 u (estimated flag)
- Rank among reporters: 6 of 132 in the stored Comtrade response

Netherlands reported primaryValue 1930661109.952 USD (shown with separators as 1,930,661,109.952) for HS 854143 imports from the world in 2024. That number is the UN Comtrade public-preview CIF primary value for partner World. Reported quantity: 22,718,291.838 u (Comtrade flagged the quantity as estimated). Money is 3 because this reporter ranks 6 of 132 reporters in that response by primaryValue. Authority is 4 because the country is where the buyers are: importers and distributors there hold the purchase order, and the country itself does not. Need is 5. The official heading names this product, so the import value is a direct Comtrade measure of recorded trade in it. Official commodity description: Electrical apparatus; photosensitive semiconductor devices, photovoltaic cells assembled in modules or made up into panels. The preview count matched the rows stored (132), under the 500-row cap.

Outreach: Do not email a country. Use Netherlands as a map: find importers there that already clear HS 854143 and open with the Comtrade fact that Netherlands reported primaryValue 1930661109.952 USD of this heading in 2024. Offer a first-container spec sheet, not a market-size story.

Source: UN Comtrade public preview API, HS 854143, 2024 imports from World - https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc

11. Pakistan - reported imports of HS 854143

- Type: Importing country
- Money: 3 / 5
- Authority: 4 / 5
- Need: 5 / 5
- Total: 12 / 15
- Link: https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc
- primaryValue USD: 2197252312.067
- primaryValue with separators: 2,197,252,312.067
- Quantity: 37417095.0 u
- Rank among reporters: 4 of 132 in the stored Comtrade response

Pakistan reported primaryValue 2197252312.067 USD (shown with separators as 2,197,252,312.067) for HS 854143 imports from the world in 2024. That number is the UN Comtrade public-preview CIF primary value for partner World. Reported quantity: 37,417,095.0 u (quantity not flagged as estimated). Money is 3 because this reporter ranks 4 of 132 reporters in that response by primaryValue. Authority is 4 because the country is where the buyers are: importers and distributors there hold the purchase order, and the country itself does not. Need is 5. The official heading names this product, so the import value is a direct Comtrade measure of recorded trade in it. Official commodity description: Electrical apparatus; photosensitive semiconductor devices, photovoltaic cells assembled in modules or made up into panels. The preview count matched the rows stored (132), under the 500-row cap.

Outreach: Do not email a country. Use Pakistan as a map: find importers there that already clear HS 854143 and open with the Comtrade fact that Pakistan reported primaryValue 2197252312.067 USD of this heading in 2024. Offer a first-container spec sheet, not a market-size story.

Source: UN Comtrade public preview API, HS 854143, 2024 imports from World - https://comtradeapi.un.org/public/v1/preview/C/A/HS?flowCode=M&period=2024&partnerCode=0&cmdCode=854143&maxRecords=500&inc

12. Alibaba.com

- Type: Marketplace
- Money: 4 / 5
- Authority: 3 / 5
- Need: 4 / 5
- Total: 11 / 15
- Link: https://www.alibaba.com/

Alibaba is a public board for module inquiries. Money is 4. Authority is 3 because some inquiries are traders. Need is 4.

Outreach: Quote a watt class you have in a warehouse and the incoterms. Ask for the project site or the distributor's company registration.

Source: Alibaba.com - https://www.alibaba.com/

13. Alibaba Sourcing

- Type: RFQ board
- Money: 4 / 5
- Authority: 3 / 5
- Need: 4 / 5
- Total: 11 / 15
- Link: https://sourcing.alibaba.com/

Alibaba Sourcing posts RFQs for modules. Money is 4 when the megawatt figure is real. Authority is 3. Need is 4.

Outreach: Bid only RFQs that name watt class and destination. A megawatt number with no site is not a bid.

Source: Alibaba Sourcing - https://sourcing.alibaba.com/

14. SEIA

- Type: Community
- Money: 4 / 5
- Authority: 3 / 5
- Need: 4 / 5
- Total: 11 / 15
- Link: https://seia.org/

SEIA (seia.org) is the U.S. solar industry association. Money is 4 because members develop and install projects. Authority is 3. Need is 4 for the U.S. market specifically.

Outreach: If the United States ranks in your Comtrade table, use SEIA to identify developers and then sell through a distributor or an EPC they already buy from.

Source: SEIA - https://seia.org/

15. SolarPower Europe

- Type: Community
- Money: 4 / 5
- Authority: 3 / 5

- Need: 4 / 5
- Total: 11 / 15
- Link: https://www.solarpowereurope.org/

SolarPower Europe (solarpowereurope.org) is the European solar industry association. Money is 4 because members include companies that buy and deploy modules. Authority is 3 because the association does not purchase. Need is 4.

Outreach: Use the membership as a map of European buyers. Write to procurement with a datasheet, not to the association secretariat.

Source: SolarPower Europe - https://www.solarpowereurope.org/

Limits

Editorial scores are judgments, not statistics. Every trade dollar in this report is a UN Comtrade primaryValue literal from the query above. BuyerRadar did not estimate a market size, a growth rate, or a buyer count. A static site cannot check a PayPal payment on a server; the button is the PayPal Smart Buttons client flow.

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